

MICROCONTROLLER-BASED CLOCK USING DS1307

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Digital wall clocks, table clocks and desk clocks with pointer or LCD display are readily available in the market. Here we present a clock that can be built in a small budget using AT89C2051 microcontroller. Additional feature of the clock is that the time display is visible even in the dark.

Circuit description

Fig. 1 shows the circuit of the microcontroller-based clock. It comprises microcontroller AT89C2051 (IC1), inverting buffer ULN2003 (IC2), real-time clock (RTC) DS1307 (IC3), regulator 7805 (IC4), non-inverting source driver UDN2982 (IC5) and a few discrete components.

Microcontroller AT89C2051 is the heart of the clock. It is an 8-bit microcontroller with 2Kb Flash programmable and erasable read-only memory (PEROM), 128 bytes of RAM, 15 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, a precision analogue comparator, on-chip oscillator and clock circuitry.

Port pins P1.7 down through P1.1 of the microcontroller are pulled up with 10k Ω resistor network RNW1 and connected to input pins 1 through 7 of inverting buffer IC2, respectively, to provide segment data for the seven-segment display. The display shows the time as 'hour.minute.' Flashing of decimal point (dp) on DIS3 indicates the seconds. The decimal points for DIS1 and DIS2 are not used here.

The selection of the four seven-segment displays is made by port pins P3.7, P3.4, P3.5 and P1.0 of the microcontroller. These pins drive the

four non-inverting buffers of driver IC5 to provide display-enable signal to the common-anode pin (either pin 3 or pin 8) of DIS1 through DIS4, respectively. If UDN2982 is directly given an unregulated 9V supply, it can support even bigger-size seven-segment displays (refer Fig. 1). The microcontroller provides the segment data and display-enable signal simultaneously in time-divide-multiplexed mode for displaying a particular number on the seven-segment display. Segment data and display-enable signal for display are refreshed every four milliseconds, so the display appears to be continuous.

Switch S1 is used to reset the microcontroller manually, while the power-on reset signal for the microcontroller is derived from the com-

bination of capacitor C3 and resistor R2. A 11.0592MHz crystal is used to generate clock frequency for the microcontroller. When the alarm time is reached, port pin P3.3 provides aural indication signal to the base of transistor T1. Transistor T1 conducts and the piezobuzzer sounds for one minute.

Timing inputs are generated by RTC DS1307. It is a low-power, real-time, full binary-coded decimal clock/calendar having 56 bytes of non-volatile static random-access memory (RAM). Address and data are transferred serially via a two-wire, bi-directional bus. The clock/calendar provides seconds, minutes, hours, day, date, month and year information. Battery connection at pin 3 provides battery backup. The battery (CR2032) works for two to three years.

The display and alarm are controlled by microcontroller AT89C2051. Data is transferred between microcontroller AT89C2051 and the RTC using two wires (which form the PC bus), one of which serves as the clock line (SCL) and the other as data line (SDA). The four subroutines required for data transfer are the send-start condition, send-stop condition, read-a-byte and write-a-byte. Using these subroutines, the time of the day can be written into internal registers of the RTC at address locations 00H through 06H (refer the datasheet). The address of the control register is 07H. If it is written with control word 10, a 1Hz square wave is available from pin 7 of the RTC. The pulse output at pin 7 drives pin 5 (dp) of DIS3, which blinks every second.

The RTC is driven by an external 32.768kHz crystal. A 3V battery is

PARTS LIST

Semiconductors:

IC1	- AT89C2051 microcontroller
IC2	- ULN2003 inverting buffer
IC3	- DS1307 real-time clock
IC4	- 7805, 5V regulator
IC5	- UDN2982 non-inverting source driver
T1	- BC547 NPN transistor
D1-D4	- 1N4007 rectifier diode
DIS1-DIS4	- LTS542 common-anode seven-segment display
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon)

R1	- 470-ohm
R2, R4, R5	- 10-kilo-ohm
R3	- 1-kilo-ohm
R6	- 470-ohm
RNW1	- 10-kilo-ohm resistor network

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic disk
C3, C6	- 1 μ F, 16V electrolytic
C4, C5	- 33pF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 9V, 500mA secondary transformer
S1, S2	- Push-to-on switch tactile switch
S3	- SPDT switch
X _{cr1}	- 11.0592MHz crystal
X _{cr2}	- 32.768kHz crystal
BATT.	- CR2032 3V lithium battery
PZ1	- Piezobuzzer

